# Expo and Defense

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

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## Expo and Defense

> Prepare Expo and Defense: check the target, prerequisites, resources and adaptable teaching schedule.

### Teach the learning cycle

Primary target: Evaluate a saved candidate and a manual **baseline** fairly

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

### Retrieval answer guidance

#### Improve One Cause

Without opening the old lesson, explain one key idea from Improve One Cause. Give an example and a mistake that the idea helps you avoid.

Look for: Repair a located defect without changing the agreed requirements

Reference example, not the only acceptable wording: Expected B1 with equality and quiet satisfied; S0 incorrectly returns no match because its time condition is strictly less than. S1 uses no more than, including equality.

Preserve validation, quiet, first-match order, advancement, stopping and reset. Save S0, its failing trace and exact request/catalogue labels beside the new S1 change record and rerun.

#### Test Normal Cases

Without opening the old lesson, explain one key idea from Test Normal Cases. Give an example and a mistake that the idea helps you avoid.

Look for: Plan routine cases with expected results derived from the requirements

Reference example, not the only acceptable wording: R1 B1 because time fits and it is quiet; early match. R2 B2 because B1 is too long, B2 fits and quiet is unrestricted; later match.

R3 no match because every time exceeds 6; exhaustion. R4 no match because B2 fails quiet while other rows exceed 9; both conditions. Record P1 version 1, B version 1 and the manual **baseline**.

A full pass needs the required **output** and a correct explanation.

### Choose Inputs and Outputs

Without opening the old lesson, explain one key idea from Choose **Input**s and **Output**s. Give an example and a mistake that the idea helps you avoid.

Look for: Turn the project outcome into precise **input** and **output** requirements

Reference example, not the only acceptable wording: Time required must be no more than available time; when quiet is required the activity must be quiet. Choose the first compatible activity in B1 to B4 order.

No means quiet is not required, not forbidden, so quiet B1 is permitted if 10 minutes fits.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### What the learner must demonstrate

> Defend a sorter with 10/12 routine successes and 2/4 mixed-material successes. Report overall and separate results, show one failure and propose a justified next step. What claim would overstate the **evidence**?

### Concept to explain

Understanding is shown by explanation, **transfer**, testing, and limits.

| Term | Meaning |
| --- | --- |
| demonstration | live showing of system behavior |
| defense | evidence-based explanation under questions |
| transfer | applying understanding to a fresh case |
| limitation | condition where performance or use is restricted |

### Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

#### Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked **demonstration** 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

### Preparation for the connected explanation

This final week evaluates the saved paper candidate, not presentation polish or an imagined trained model. Fix instructions before revealing the new catalogue and requests.

Keep the key below teacher-only until **observation**s are recorded. If a pupil has already seen these cases, use a fresh teacher-checked set and record that substitution; do not call familiar cases unseen.

### Readiness answer

No. It informed the change and is now development **evidence** for the revision. Keep the old candidate and its final result; a fresh evaluation of the revision needs new reserved cases.

Saved candidate, manual **baseline**, development folder and blank record sheets with the fields listed in W2. Provide the catalogue and requests without the answer key after candidates are fixed.

Reserved catalogue B version 4 keeps B1 to B4 order and names, with times 18, 14, 11, 7 minutes and quiet labels Yes, No, Yes, No. All materials are assumed available. Same **input** and matching rules as before.

Reserved requests and key: F1 7 No → B4; F2 11 Yes → B3; F3 14 No → B2; F4 18 Yes → B1; F5 6 No → no match; F6 7 Yes → no match; F7 0 No → no match; F8 30 Yes → B1; F9 31 No → clarify; F10 blank minutes Yes → clarify.

For F1–F4, earlier rows fail time; the selected row fits with equality and satisfies quiet. F5 and F7 fail all times. F6 fits only nonquiet B4, which quiet excludes. F8 accepts first quiet B1.

F9 and F10 fail validation before search.

**End goal for the pictorial case**

By the end, I can defend a project using measured results, limits and an honest **baseline** comparison.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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TEACHER GUIDE · 2 OF 9

## Explain the mechanism

> Model Expo and Defense using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

### 1. Input — information entering

**INPUT** fresh item and documented system

### 2. Process — the deciding step

**PROCESS** execute and explain each stage under questions

### 3. Output — the result

**OUTPUT** result, **evidence**, **limitation**, and responsible next action

### Key idea

CORE IDEA Understanding is shown by explanation, **transfer**, testing, and limits.

### How the mechanism works

UNDER THE HOOD A **defense** evaluates **transfer**able understanding, not presentation polish.

Strong **evidence** includes system architecture, version history, planned tests, failures, controlled improvement, fairness/privacy boundaries, and independent response to a novel case.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

### The case to investigate

CASE A team presents the Classroom Sorter, traces a new item live, shows test **evidence**, explains one failure, and states when a human must decide.

### Step 1

STEP 1 — OBSERVE fresh item and documented system

### Step 2

STEP 2 — TRACE execute and explain each stage under questions

### Step 3

STEP 3 — CONCLUDE result, **evidence**, **limitation**, and responsible next action

### Model answer

MODEL REASONING A complete **defense** traces **input**, **process**, and **output**; distinguishes rules from learned parts; presents normal and edge tests; explains one controlled improvement; states privacy, fairness, uncertainty, and human-review boundaries; and handles a fresh case independently.

### A claim to avoid

DO NOT OVERCLAIM Confidence, decoration, or memorized vocabulary cannot replace a fresh trace and **evidence**-based answers.

What new **evidence** could change this answer?

**Concrete teaching case**

### The teaching case

The expo poster says always correct after six classroom passes.

### The teaching case

Write a defensible claim.

**Exact explanation to model**

> ### Why the result follows
>
> Report six specified passes with conditions and limits. Show an **input**, decision trace, **output** and test record, including an uncertain case.

**Explain the added worked example**

## Explain the added worked example · 1

This week you will evaluate your activity adviser and explain what its **evidence** supports. A project with an honestly reported failure can show strong understanding.

Save your candidate instructions, version label, **input** rules, development records and known problems before your teacher reveals the final requests and catalogue. Keep the instructions unchanged during this evaluation.

## Explain the added worked example · 2

The agreed rule remains: validate whole available minutes from 0 to 30 and Quiet required Yes or No. No allows quiet or nonquiet activities.

Start at the first row and suggest the first activity whose time is no more than available time AND whose quiet label satisfies the request. Explain the choice. If all rows fail, report no match.

Invalid or missing **input** needs clarification before searching. Clear the previous **output** for each run.

## Explain the added worked example · 3

Your teacher will supply a fresh, labelled catalogue in the same format. Its values are test **input**s, not permission to rewrite your instructions. All materials are assumed available.

The manual **baseline** checks every row against the same requirements, then selects the first fit. Both methods advise only; neither books, publishes or chooses on a person’s behalf.

## Explain the added worked example · 4

Use the same final requests, catalogue, requirements and full-pass rule for both methods. Record the candidate **output** and reason before looking at the expected answer.

Record the **baseline** separately; do not copy the candidate result into it. Your teacher holds the answer key.

A full pass needs both the required **output** and a valid reason, including why earlier rows fail when a later row is selected.

## Explain the added worked example · 5

For each request record its ID, **input**s, catalogue label, candidate label, candidate **output** and reason, **baseline output** and reason, and both results after checking. Keep a trace for any failure. Mark unrun cases explicitly.

Do not silently correct an **observation**. If a person misreads an instruction, record that execution error separately from a defect in the written instruction.

### Key terms

| Term | Meaning |
|---|---|
| Final evaluation | Testing a saved candidate under stated conditions after development |
| Baseline | A comparison method tested under the same conditions |
| Evidence | Recorded observations that support or challenge a claim |
| Limitation | A boundary on what a result or system can establish |
| Defence | An explanation of a decision using evidence and reasoning |

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

## Model the reasoning aloud · 6

Here is a fictional example from a different project, not your class results. A candidate has 6 full passes from 8 attempted cases; the **baseline** has 8 from the same 8. Two more planned cases were not run.

The candidate passed 75 percent of attempted cases; the **baseline** passed 100 percent. Neither has **evidence** for the two unrun cases. The candidate did not outperform the **baseline** on this measure.

| Evidence | Supported statement | Claim to avoid |
|---|---|---|
| 6 of 8 full passes | Six attempted cases met the full rule | It works for every request |
| Both methods pass the same cases | They tie on this measured result | Automation must be better |
| A time was not measured | Speed is unknown | The candidate is faster |
| Only fictional catalogue data | Results concern these test conditions | People will enjoy every suggestion |

## Model the reasoning aloud · 7

Explain each failure and its consequence instead of hiding it inside a total. Even perfect results on a small reserved set do not establish reliability in every situation.

Our paper adviser applies human-**written rule**s; it has not learned a model from **training** examples. It checks compatibility, not enjoyment, popularity or real material availability.

> **In the fictional example, state the candidate and baseline results, explain how the two unrun cases should be reported, and give one useful next action. Why is “the adviser is ready for every classroom” unsupported?**

| Thinking move | Teacher prompt |
|---|---|
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Expo and Defense

A fictional paper sorter team prepares an exhibition **evidence** board.

### The facts we will use

#### Supplied facts

Both prototype and manual **baseline** attempt the same 10 reserved cases. Prototype passes 8; **baseline** passes 9.

#### Supplied facts

Two additional planned cases are unrun. No timing or pupil-satisfaction measures were collected.




Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/36

# Expo and Defense

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Describe contract

Show the task, inputs, rule and reserved case selection.



## 2. Show results

Prototype 8/10=80%; baseline 9/10=90% on attempted cases.



## 3. Keep incomplete work

Two unrun cases remain visible, without pass/fail labels.



## 4. Defend conclusion

Prototype did not beat baseline on this measured outcome; explain its failures.

**CHANGE ONE THING**

Someone adds the two unrun cases to the denominator as failures.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Describe contract | Show the task, inputs, rule and reserved case selection. |
| 2. Show results | Prototype 8/10=80%; baseline 9/10=90% on attempted cases. |
| 3. Keep incomplete work | Two unrun cases remain visible, without pass/fail labels. |
| 4. Defend conclusion | Prototype did not beat baseline on this measured outcome; explain its failures. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

### Why the result follows

An exhibition should make the **evidence** inspectable. A polished **demonstration** is not a substitute for the reserved-case record.

The shared cases allow a direct descriptive comparison of pass rates; missing timing data prevents a claim that the prototype is faster. Unrun cases must not be treated as successes.

### What this example does not establish

#### Keep the stated limit

This fictional evaluation does not demonstrate classroom learning impact or performance on every future request.

### Change one thing in the pictured case

#### Change one thing

Change: Someone adds the two unrun cases to the denominator as failures.

#### Predicted result

**Prediction**: That reports a different measure, 8/12 completed successes out of planned cases.

#### Why the result follows

Reason: It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

### Ask while showing the picture

## Ask while showing the picture · 8

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

# Demonstrate and guide practice

> Demonstrate Expo and Defense, reduce support deliberately and check individual reasoning before discussing answers.

## Board sequence

### 1. Put the facts on the board

Write the given facts: The expo poster says always correct after six classroom passes.

### 2. State the decision to explain

Write the question to explain: Write a defensible claim.

### 3. Reveal the worked reasoning

Reveal after **prediction**s: Report six specified passes with conditions and limits. Show an **input**, decision trace, **output** and test record, including an uncertain case.

### Hold back the independent task

Keep for an independent attempt: At the expo, a team says our sorter never fails after testing four cards. Replace the claim with an **evidence**-based statement and name a next test.

### Classroom dialogue

#### Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

#### Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

### Model these guided questions

Fictional candidate 6/8 and **baseline** 8/8 attempted, with two unrun planned cases each; candidate 75 percent and **baseline** 100 percent on that set. Investigate retained failure traces before a versioned repair.

Neither has **evidence** for all classrooms. Exit answers must use the pupil's real record, limit the conclusion, name a next step and distinguish explicit rules from fitted models. Passing tests grants no booking authority.

### Case-specific teaching sequence

| Stage | Teacher action |
|---|---|
| Session 1 start to 7 minutes | Check the prerequisite and W1; collect or photograph each fixed instruction set before revealing any final inputs. |
| 7 to 12 minutes | Reveal B version 4 and F1–F10 requests without the key. Explain W2 record fields and the unchanged full-pass criterion. |

| Stage | Teacher action |
| --- | --- |
| 12 to 37 minutes | Run all ten cases with candidate and baseline. Partners swap operating and recording; preserve each output before comparison with the key. Record assistance and unrun cases. |
| 37 to 45 minutes | Reveal the key, check reasons and summarise actual results in W2. Retain failure traces; do not edit the evaluated instructions. |
| Session 2 start to 8 minutes | Discuss the fictional 6-of-8 example and collect the guided explanation. Keep hypothetical results separate from class records. |
| 8 to 22 minutes | Prepare W3 and run paired two-minute presentations with questions and role swaps. Sample individual traces; offer accessible spoken or written responses. |
| 22 to 37 minutes | Collect W4 independently before hints. Assess evidence claims and the meaning of a rerun after revision. |
| 37 to 45 minutes | Complete W5 and the exit. Archive the candidate, final results and next action; note support needs for Grade 7. |

### Reduce help deliberately

#### Completion practice

Completion task: Use a separate sheet for the teacher's reserved cases. For every case record its ID and **input**s, catalogue and candidate labels, both methods' observed **output**s and reasons, and results after checking.

Preserve failure traces. Here summarise each method's full passes over attempted cases and list unrun cases. State what the comparison supports.

#### Reasoning to listen for

Teacher check: Use the reserved key in Preparation only after **observation**s are saved. Full pass requires **output** and reason; no-match explanations must rule out all rows. Clarification must precede search.

Report actual candidate and **baseline** counts over the same attempted cases, plus unrun cases. Never substitute the key for an **observation** or assume a **baseline** run was correct.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

### Individual reasoning check

Use the worked case for Evaluating and explaining your project. Proposed explanation: “A polished presentation proves the system works”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **Expected repair**
>
> Expected correction: Ask for the saved instruction, trace and actual result behind the claim.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

### Pictorial practice question

Write one defensible success statement and one claim the **evidence** cannot support.

### Reasoned teacher answer

Defensible: 8 of 10 attempted reserved cases passed. Unsupported: the prototype is faster or more accurate than the **baseline**.

Changed-case reasoning: That reports a different measure, 8/12 completed successes out of planned cases. It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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TEACHER GUIDE · 4 OF 9

## Run the weekly evidence lab

> Run the Expo and Defense investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

### 1. Predict

State your **prediction** before the result is known.

### 2. Investigate

Complete the task: Demo the system, solve a fresh case, and answer why it may fail.

### 3. Compare cases

Keep a normal case and an edge case clearly separated.

### 4. Record honestly

Record what happened—even when it does not match your **prediction**.

### 5. Explain and plan the next test

Explain the result, one **limitation**, and the next useful test.

### Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

#### Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

| Record | What to capture |
| --- | --- |
| Input or evidence | Exact data, instruction or claim |
| Expected result | Answer or safe fallback declared before the run |
| Observed result | Actual result, including errors |
| Explanation | Rule, calculation or evidence explaining the outcome |
| Next test | One justified change and a new prediction |

#### Diagnose and reteach

### Find the unsupported claim

CLAIM TO REPAIR A successful **demonstration** proves universal reliability.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

### Address these additional misconceptions

A polished presentation proves the system works: Ask for the saved instruction, trace and actual result behind the claim.

Unrun cases count as passes: Report attempted denominators and unrun cases separately.

The **baseline** must lose: A tie or a better **baseline** can justify retaining the simpler method.

A repair erases the final failure: Preserve the evaluated version and results; revealed cases become development checks for the revision.

### Match the support to the first error

| What the response shows | Next teaching action |
| --- | --- |
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

# Assess and explain the answers

> Use the Expo and Defense questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

### Worked case answer

#### Teacher key · Worked case answer

Report six specified passes with conditions and limits. Show an **input**, decision trace, **output** and test record, including an uncertain case.

### Independent question

#### Independent question · withhold the key

At the expo, a team says our sorter never fails after testing four cards. Replace the claim with an **evidence**-based statement and name a next test.

### Expected independent answer

#### Teacher key · Expected independent answer

Say it succeeded on these four named cards, with the observed count reported. Four cases do not prove it never fails. Next test unfamiliar, blurred or dim examples and explain human review.

### What counts as evidence

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Result | Incorrect or absent | Partly correct | Correct with necessary conditions |

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Reasoning | No supporting trace | Some steps | Clear mechanism, calculation or source support |
| Limit | Overclaims certainty | Vague limitation | Relevant uncertainty or missing fact |
| Transfer | Copies without adapting | Adapts with prompts | Solves a changed case independently |

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

### Feedback

Ask the learner to find the **evidence** that challenges this claim: “A successful **demonstration** proves universal reliability.” Keep the first attempt and an uncoached retry separate.

### Original transfer answer

#### Teacher key · Original transfer answer

Overall is 12/16=75%; routine is about 83.3%; mixed is 50%. Show the actual failed trace and investigate mixed-material handling or review rules. 'Works reliably on every material' overstates these small specified tests.

Advancement requires independent explanation, not presentation polish alone.

### Explanations for the additional workbook examples

### W1 Record the fixed candidate

#### Workbook question · W1

Before final cases are revealed, record your version label, where its complete instructions are saved, the agreed **input** and matching rules, the manual **baseline** and any known **limitation**.

Confirm that the instructions stay unchanged while the assigned catalogue is used as **input**.

### Teacher reasoning · W1

Require an identifiable saved version, complete instructions, valid whole minutes 0 to 30, Yes or No quiet **input**, inclusive time AND quiet compatibility, first-fit order, reset, stopping and clarification.

The **baseline** checks all rows and selects the first fit under the same contract. Accept honestly stated known failures; do not award a stronger result for hiding them.

## W2 Keep the final evaluation record

### Workbook question · W2

Use a separate sheet for the teacher's reserved cases. For every case record its ID and **input**s, catalogue and candidate labels, both methods' observed **output**s and reasons, and results after checking. Preserve failure traces.

Here summarise each method's full passes over attempted cases and list unrun cases. State what the comparison supports.

### Teacher reasoning · W2

Use the reserved key in Preparation only after **observation**s are saved. Full pass requires **output** and reason; no-match explanations must rule out all rows. Clarification must precede search.

Report actual candidate and **baseline** counts over the same attempted cases, plus unrun cases. Never substitute the key for an **observation** or assume a **baseline** run was correct.

### W3 Prepare your evidence defence

#### Workbook question · W3

Write a two-minute explanation of the problem, one trace, your actual comparison, a **limitation** and a next action. Explain why this is a written-rule design rather than a model trained on examples.

State which personal information is unnecessary and who decides whether to act.

#### Teacher reasoning · W3

Look for a saved trace connected to the actual instruction, accurate counts, a specific failure or **limitation** and a justified next step. The project applies explicit human-**written rule**s, with no fitted model.

Only minutes and quiet preference are needed; names and reasons for quiet are unnecessary. The person may decline and the teacher checks real conditions; the adviser does not book or publish.

### W4 Judge a different project independently

#### Workbook question · W4

A fictional library adviser and its manual **baseline** face the same 6 reserved cases. The candidate has 4 full passes; the **baseline** has 5. One additional planned case is unrun by both. No speed data was collected.

The team says “ours is faster and more reliable everywhere” and repairs it after seeing the failures. Explain what the counts support, two unsupported claims and how to label a rerun of the revealed cases.

### Teacher reasoning · W4

Report candidate 4 of 6 and **baseline** 5 of 6 full passes, with one unrun case for each. The **baseline** performed better on this measured set; neither result establishes reliability everywhere. Speed was not measured.

A rerun after the failures informed a repair is development **evidence** for the revised version, not an unseen final test. Preserve the old results and label both versions.

**W5 Write your final conclusion**

### Workbook question · W5

Use your own records to complete: Under these conditions, our saved version showed… The **baseline** showed… We still do not know… Our next action is… Include one privacy or human-control choice.

If a case failed, explain its consequence without deleting the result.

### Teacher reasoning · W5

Accept different conclusions grounded in actual **observation**s, including a decision to retain the **baseline**. Require stated conditions, honest comparison, a real uncertainty and a specific action.

A quiet requirement does not require a diagnosis; an **output** does not authorise action. A failure remains part of the evaluated version’s record even after a repair.

**Assess independent transfer**

### Independent question · withhold the key

Selected case: A fictional library adviser and its manual **baseline** face the same 6 reserved cases. The candidate has 4 full passes; the **baseline** has 5. One additional planned case is unrun by both. No speed data was collected.

The team says “ours is faster and more reliable everywhere” and repairs it after seeing the failures. Explain what the counts support, two unsupported claims and how to label a rerun of the revealed cases.

### Selected case answer

#### Teacher key · Selected case answer

Report candidate 4 of 6 and **baseline** 5 of 6 full passes, with one unrun case for each. The **baseline** performed better on this measured set; neither result establishes reliability everywhere. Speed was not measured.

A rerun after the failures informed a repair is development **evidence** for the revised version, not an unseen final test. Preserve the old results and label both versions.

| Dimension | Evidence required |
|---|---|
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

#### Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

### Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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## Connect and extend learning

> Connect Expo and Defense to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

### Helpful Classroom Sorter

Present the final Classroom Sorter portfolio and complete the AI-1 certification **defense**.

Keep the decision, **input**, result, explanation and remaining **limitation** in the project **evidence** record. Retrieve this record next week rather than restarting the project.

### Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

### Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

### Bridge to the next week

Review the full project **evidence** and choose concepts needing another independent attempt before the next level. One expo is not proof of mastery of all AI.

### Support and fresh reassessment

Provide a blank record grid, read instructions aloud and allow oral **defence**, while recording assistance. Keep independent W4 responses before feedback.

Allow additional time for unrun cases with the same fixed version and record when they ran; do not manufacture completion.

Extension: propose an additional test for a stated **limitation** and explain why it would help without claiming proof of every situation.

Assess four strands separately: faithful execution with traces; honest comparison including failures; independent explanation of rule-based **mechanism** and **human control**; justified limits and next action.

Mark each secure, developing or not yet **evidence**d, with one cited **observation**. Do not infer individual mastery from group totals. W4 requires 4/6 versus 5/6, one unrun, unknown speed and a development rerun.

After support, retry with candidate 3/5 and **baseline** 3/5, two unrun and no timing: a tie on observed passes, no universal or speed claim. Require fresh reasoning.

Grade 6 Foundations is complete. Grade 7 AI 2 begins programming and data. Carry forward versioned instructions, validation, tests, privacy and **evidence**-based conclusions; use the recorded assessment strands to plan support.

### Plan the delayed return

Suggested return points in this level: plan a review after the level ends. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
|---|---|
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |

| Record | Teacher evidence |
| --- | --- |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

Present the Helpful Classroom Sorter's versions, test records, **baseline**, failures and next investigation as an **evidence** portfolio.

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## Prepare, teach and assess

> Use this preparation summary alongside the detailed Expo and Defense lesson; it is a checklist rather than a replacement for the full sequence.

### Teaching target

Teaching target: Evaluate a saved candidate and a manual **baseline** fairly

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Use a separate sheet for the teacher's reserved cases.

For every case record its ID and **input**s, catalogue and candidate labels, both methods' observed **output**s and reasons, and results after checking. Preserve failure traces.

Here summarise each method's full passes over attempted cases and list unrun cases. State what the comparison supports.

### Supported practice answer

Reasoned practice answer: Use the reserved key in Preparation only after **observation**s are saved. Full pass requires **output** and reason; no-match explanations must rule out all rows. Clarification must precede search.

Report actual candidate and **baseline** counts over the same attempted cases, plus unrun cases. Never substitute the key for an **observation** or assume a **baseline** run was correct.

Check every learner before discussion: Use the worked case for Evaluating and explaining your project. Proposed explanation: "A polished presentation proves the system works". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Repair the misconception

Misconception repair: Ask for the saved instruction, trace and actual result behind the claim.

### Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

### Extend the reasoning

Extension: Someone adds the two unrun cases to the denominator as failures. Require a **prediction** and a reason before comparing. Expected result: That reports a different measure, 8/12 completed successes out of planned cases.

Reason: It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

### Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

### Project checkpoint

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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## Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

### For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and **limitation**s. Progress ticks and quiz selections alone do not establish mastery.

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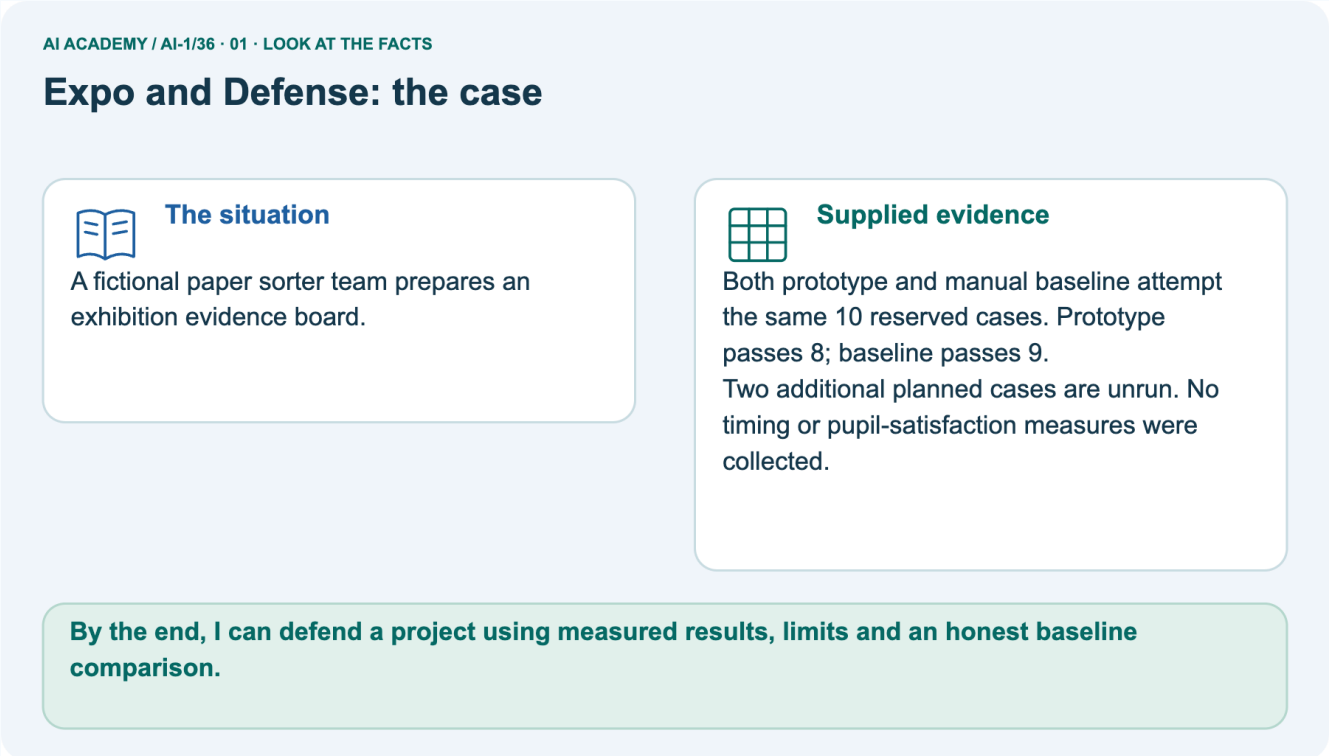
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## Visual concept journey

> Show one Expo and Defense graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

### 01. Expo and Defense: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional paper sorter team prepares an exhibition **evidence** board. Supplied facts: Both prototype and manual **baseline** attempt the same 10 reserved cases. Prototype passes 8; **baseline** passes 9.

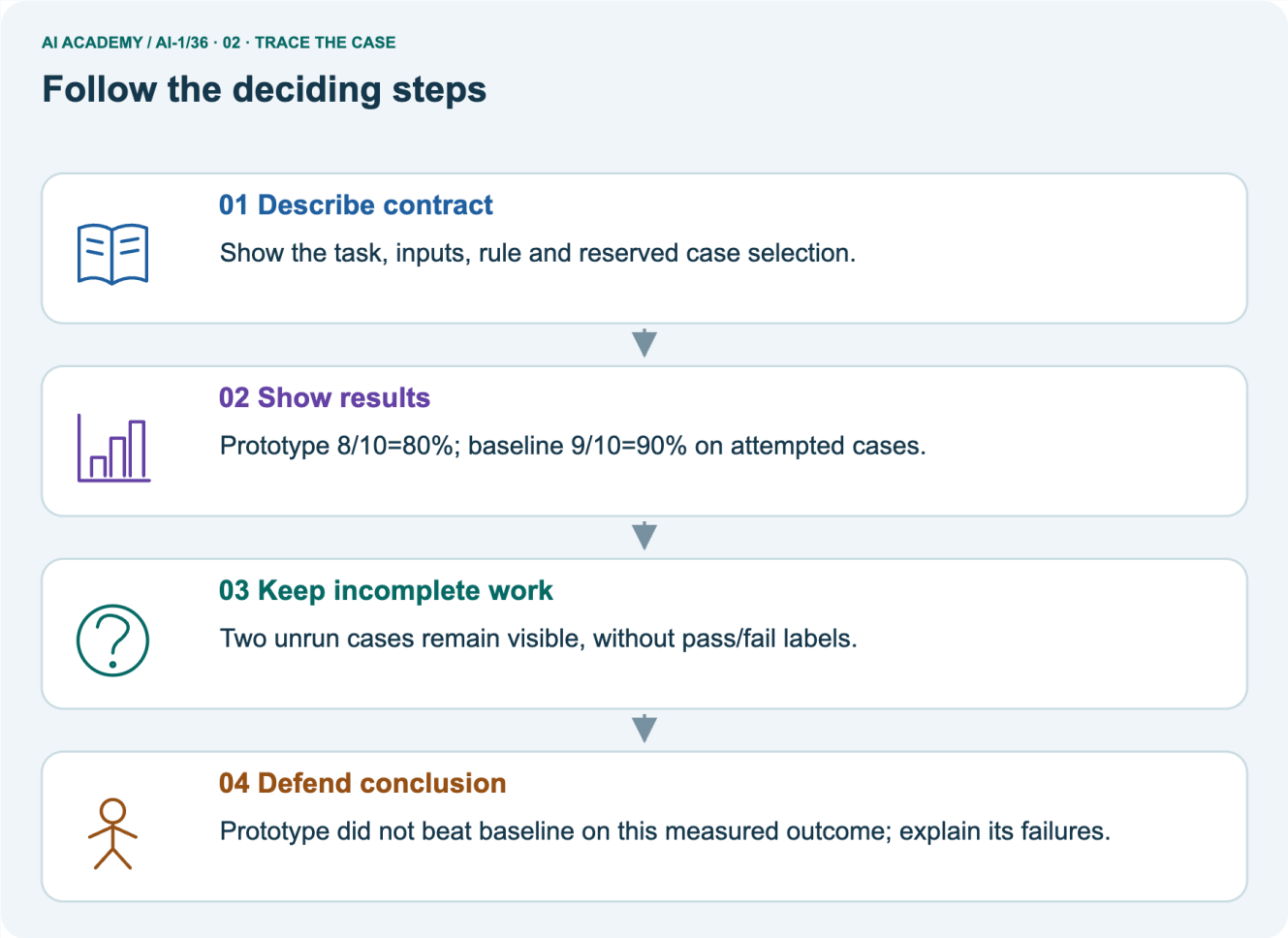
Two additional planned cases are unrun. No timing or pupil-satisfaction measures were collected.

Point to the given fact that matters most. Explain why it matters.

### Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

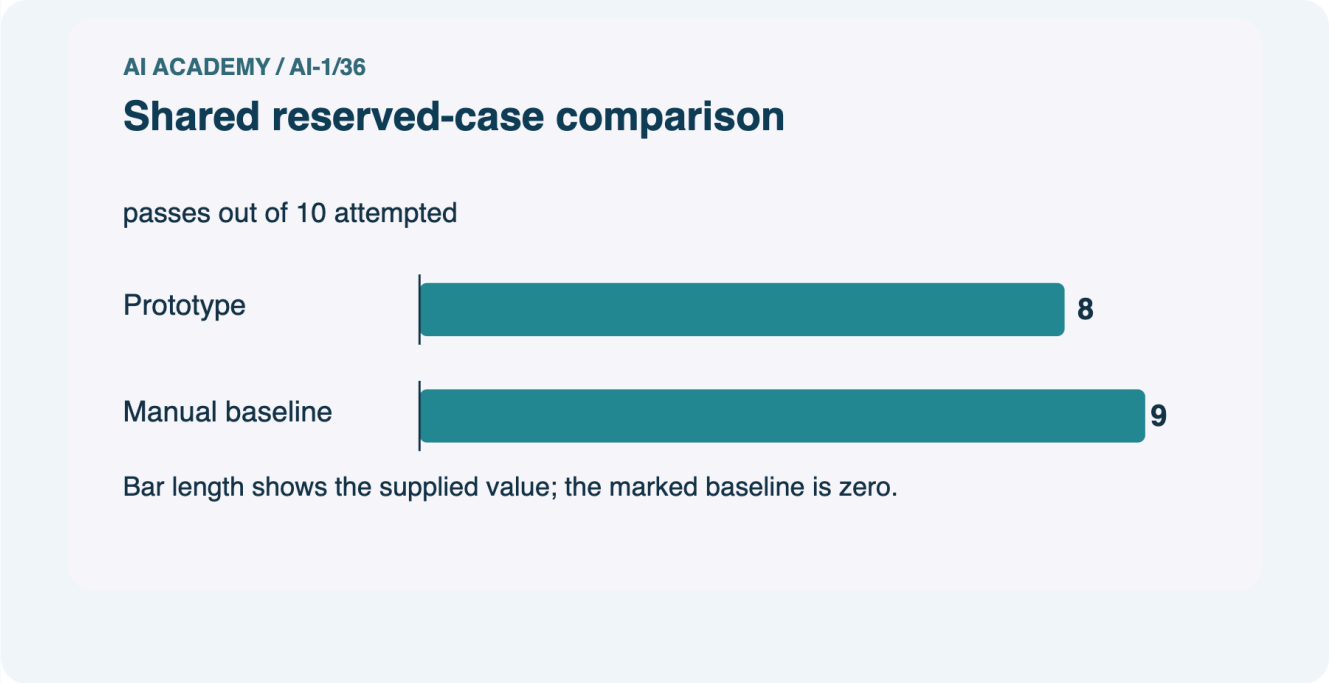
Describe contract: Show the task, **input**s, rule and reserved case selection. → Show results: Prototype 8/10=**80%**; **baseline** 9/10=90% on attempted cases. → Keep incomplete work: Two unrun cases remain visible, without pass/fail labels. → Defend conclusion: Prototype did not beat **baseline** on this measured outcome; explain its failures.

> Point to each numbered stage and explain which recorded **input** it uses.

### Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

### 03. Shared reserved-case comparison




Follow the labelled picture with the matching explanation. Open the vector diagram

Shared reserved-case comparison. Units: passes out of 10 attempted. Prototype: 8. Manual **baseline**: 9.

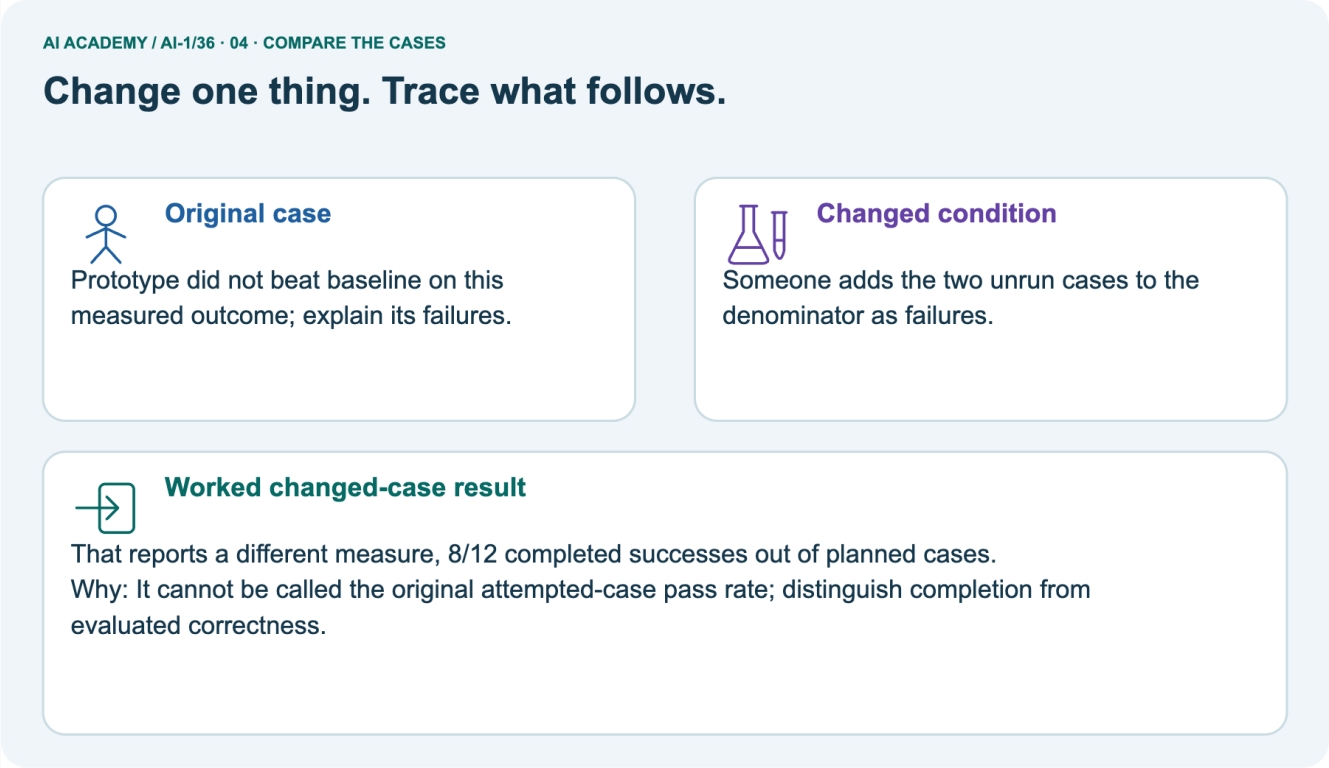
Read the labels and values. Explain the relationship the model shows.

### Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Prototype did not beat **baseline** on this measured outcome; explain its failures.
Changed condition: Someone adds the two unrun cases to the denominator as failures.

Expected result: That reports a different measure, 8/12 completed successes out of planned cases.
Reason: It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

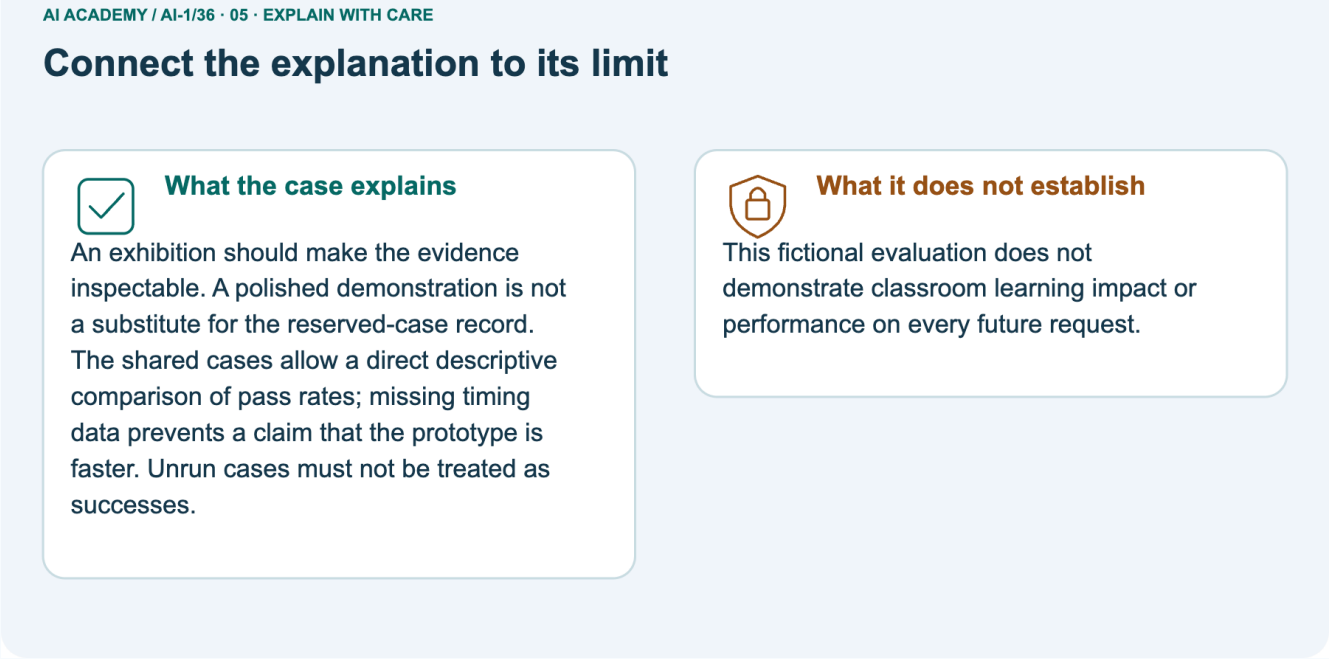
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

An exhibition should make the **evidence** inspectable. A polished **demonstration** is not a substitute for the reserved-case record.

The shared cases allow a direct descriptive comparison of pass rates; missing timing data prevents a claim that the prototype is faster. Unrun cases must not be treated as successes.

Limit: This fictional evaluation does not demonstrate classroom learning impact or performance on every future request.

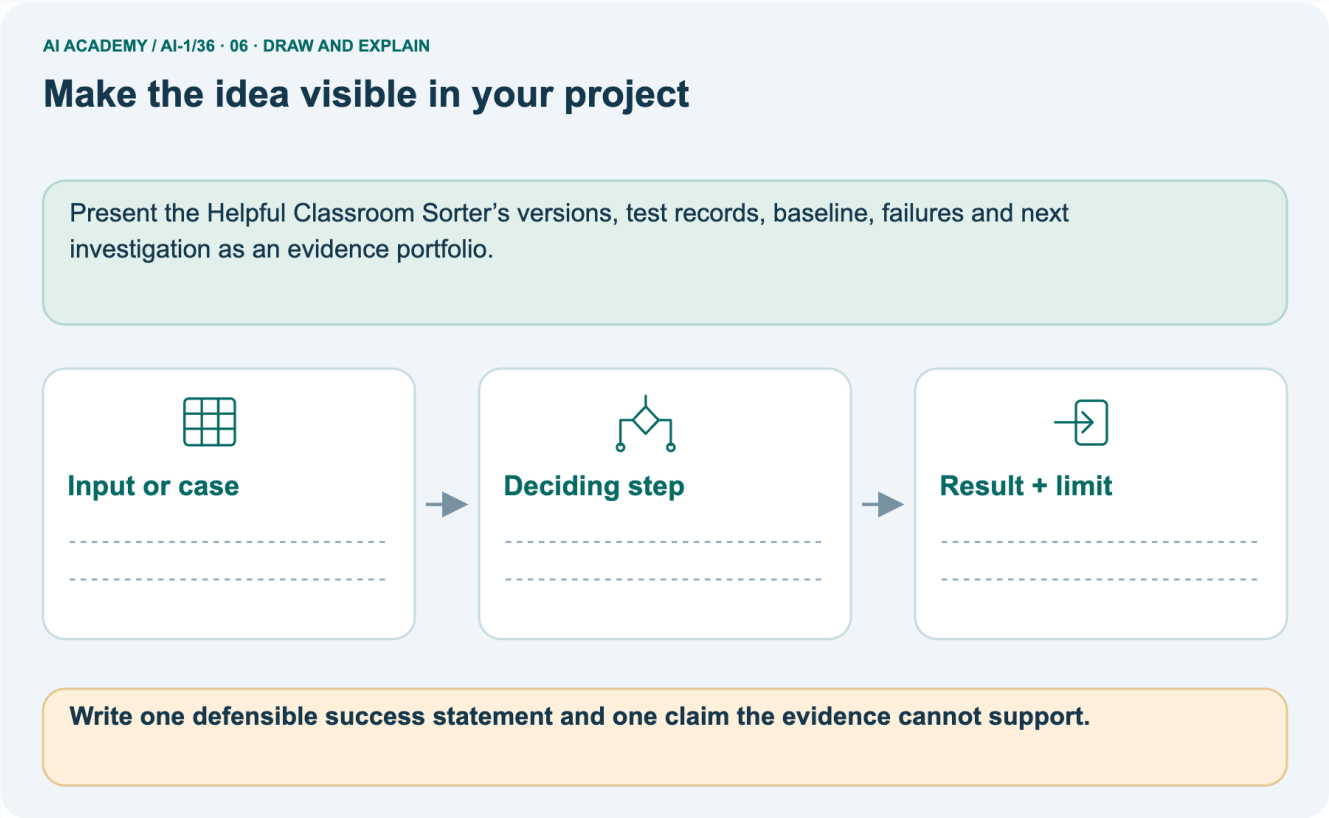
Explain the case, then point to a limit on the claim.

### Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This fictional evaluation does not demonstrate classroom learning impact or performance on every future request.

### 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Present the Helpful Classroom Sorter's versions, test records, **baseline**, failures and next investigation as an **evidence** portfolio. Draw your **input**, deciding step, result and limit.

Supported question: Write one defensible success statement and one claim the **evidence** cannot support.

> Draw your project trace and justify the deciding step using **evidence**.

#### Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: Defensible: 8 of 10 attempted reserved cases passed.
> Unsupported: the prototype is faster or more accurate than the **baseline**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.